Three-dimensional atom localisation with calibrated uncertainty from multislice electron ptychography: mapping polar order in a $PbTiO_3/SrTiO_3$ superlattice

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The polarisation within a ferroelectric polar vortex rotates in three dimensions, so imaging one along a zone axis integrates away the component parallel to the beam and removes part of the structure it is made to observe. Multislice electron ptychography recovers depth from a single orientation, but its single-atom response is roughly an order of magnitude more extended along the beam than across it, so an oxygen atom's response overlaps that of the titanium 1.95 Å away. A 4D-STEM experiment on a relaxed $PbTiO_3/SrTiO_3$ supercell was simulated and reconstructed, and every atom located by modelling the volume as a superposition of a measured single-atom response. The method recovers the overlapped oxygen that deconvolution followed by peak detection does not, and attaches per-atom uncertainties combining a joint Cramér–Rao bound with Mondrian conformal calibration, which hold their nominal coverage on held-out atoms and carry over to an independent reconstruction without a tuned constant. Propagated through the Ti–O_6 off-centring, these recover the in-plane polarisation to a median 0.007 Å and 0.9°, while establishing without reference to ground truth that the along-beam component is not determined. Core-hole density functional theory indicates that the missing component remains spectroscopically accessible, although what limits the measurement is the probe convergence rather than the collection aperture.

I. INTRODUCTION

This project investigates the three-dimensional structure of polar vortices in a $PbTiO_3/SrTiO_3$ (PTO/STO) superlattice. It combines multislice electron ptychography (MEP) with calibrated atom picking to map the material. The limits of measuring the along-beam polarisation are tested, then first-principles electron energy-loss spectroscopy (EELS) simulations are carried out to investigate if the missing structural information can be recovered spectroscopically.

Ferroelectric oxide superlattices have attracted a great deal of attention over the past decade due to their unique polarisation topologies that have no counterpart in bulk ferroelectrics. Ordered arrays of vortex-antivortex pairs were first resolved in PTO/STO superlattices [1], and room-temperature polar skyrmions were later found in the same system [2]. A wider family of related topological phases has since been catalogued [3]. In these superlattices the competition between elastic, electrostatic and gradient energies drives the local dipoles into vortices [3], and it is the topology of these that is of interest. To measure this topology experimentally means measuring a three-dimensional vector field at atomic resolution.

The standard tool used for this is atomic-resolution scanning transmission electron microscopy (STEM) to measure the oxygen and B-site displacements and calculate the polarisation vectors. However, an image formed this way is a projection through the specimen, and so returns only the component of the off-centring that lies in the image plane. For a uniformly polarised domain this is a minor restriction, and can be lifted by imaging a second zone axis. For a vortex it cannot, because the polarisation rotates with depth. Different depths within a single atomic column then carry different, and sometimes opposing, dipoles, and the image reports their average.

Multislice electron ptychography (MEP) provides a solution. A conventional annular detector sums the scattering over a range of angles as well as along the beam, discarding both. MEP instead records the full diffraction pattern at every scan position on a pixelated detector. By modelling dynamical scattering through a stack of transmission slices, this recovers the phase of the scattered electron wave, mathematically resolving depth from a single orientation as first demonstrated by Gao *et al.* [4]. Because its lateral resolution is set by the illumination angle rather than by the lens [5], ptychography has demonstrated the current highest lateral resolution of any imaging method [6, 7], reaching deep sub-ångström scales even on uncorrected instruments [8]. For a single scanning pass the lateral and axial resolutions scale as $\lambda/2\alpha$ and λ/α^2 respectively, which for the 300 keV ($\lambda = 1.969$ pm), 100 mrad probe used here give 0.10 Å and 2.0 Å. MEP has been successfully applied to measure dopant depths [7] and locate oxygen vacancies [9] so should be usable for this use case. Coupling a few small-angle projections sharpens the depth resolution further, to below a nanometre [10], at the cost of no longer working from one orientation. Finally, while atomic electron tomography can also provide three-dimensional coordinates [11], it requires a large tilt series and hence a much higher beam dose, and is geometrically impractical on epitaxial thin samples.

The difficulty lies in what happens after the reconstruction. Because the reconstructed phase object is blurred along the beam, features closer than the 2.0 Å axial resolution limit merge together in depth. Viewed down this zone axis, an oxygen atom and its titanium neighbour are separated by just 1.95 Å axially, placing them right at this limit. Picking out the lighter oxygen from the blur of the heavier titanium is difficult, but possible provided they have some small lateral displacement from the ferroelectric off-centring. Simple local peak detection, however, still fails to separate the two. The established remedy of three-dimensional Richardson–Lucy or maximum-entropy deconvolution [12] was developed for sparse features, rather than for a dense sublattice. Instead, the accepted two-dimensional approach is model-based fitting of a superposition of peaks [13], which handles overlap explicitly. Carrying it into the third dimension needs a kernel for the axial response. Modelling that response as a Gaussian already places isolated dopants in depth [7]. On a dense lattice the neighbours overlap, so the tails of the kernel matter as much as its width, and it is measured here rather than assumed: propagating a single atom through the same simulation and reconstruction returns it directly.

A second difficulty is uncertainty quantification. Because a polarisation map is built from sub-ångström differences between atomic coordinates, any measured off-centring cannot be physically interpreted without a calibrated per-atom uncertainty. Standard independent error bounds assume neighbouring atoms are known exactly, which is not true here.

This report tackles both difficulties directly. A 4D-STEM experiment on a relaxed PTO/STO supercell was simulated and reconstructed by MEP. Fitting the reconstructed phase volume as a superposition of a measured single-atom response extended model-based quantification [13] to three dimensions for all species simultaneously. Every atom then carries its own uncertainty, built from a joint Cramér–Rao bound and a conformal calibration. Those intervals were shown to transfer to an independent reconstruction, and were propagated into a map of the Ti–O_6 off-centring. Finally, core-hole density functional theory (DFT) was used to determine if electron energy-loss spectroscopy (EELS) can recover the along-beam polarisation component that ptychography struggles with. No ground-truth coordinate enters the detection or the fitting, but the atom finder is not assumption-free: it is told that the material is a perovskite with three species, whose columns are pure lead, pure oxygen, or alternating titanium and oxygen. The known structure enters twice, to fit the conformal quantiles and to score the result.

II. METHODS

A. Model system

The test object was a relaxed atomistic supercell of a PTO/STO superlattice provided by the collaborators named in the acknowledgements. The cell is $18 \times 18 \times 12$ unit cells, $70.0 \times 70.0 \times 47.9$ Å, stacked along the short axis as three $SrTiO_3$, six $PbTiO_3$ and three $SrTiO_3$. The beam was set along [100], one of the 70 Å axes, so it runs along the layers rather than through them. The 20 Å scan window sits inside a single ≈ 24 Å $PbTiO_3$ block, so everything imaged in the report is $PbTiO_3$ with no $SrTiO_3$. The $SrTiO_3$ enters only through the strain and the polar texture it causes. Both are perovskites: Pb or Sr on the A site, Ti on the B site inside an oxygen octahedron. In the $PbTiO_3$ layers both cations shift off centre and the dipoles order into a labyrinth pattern. The local polarisation was quantified throughout by the B-site off-centring,

$$\boldsymbol{\delta}_i \;=\; \mathbf{r}_{\mathrm{Ti},i} \;-\; \tfrac{1}{6}\sum_{j\in \mathrm{O}_6(i)} \mathbf{r}_{\mathrm{O},j}, \tag{1}$$

the displacement of each titanium from the charge centre of its six nearest oxygens. Figure 1(a) maps it over the whole cell.

The system is a hard test of localisation. The scattering contrast spans two orders of magnitude, from lead ($Z = 82$) to oxygen ($Z = 8$), placing the light sublattice near the noise floor, and the signal of interest is a sub-ångström displacement of atoms 1.9–3.9 Å apart. The ≈ 70 Å beam path runs along a direction over which the polarisation twists into a vortex tube within the scanned volume (Fig. 1(c)).

B. 4D-STEM simulation

A convergent probe with a semi-angle of $\alpha = 100$ mrad at 300 keV was focused 20 Å above the sample entrance, giving the $\approx$4 Å diameter probe drawn on the column lattice in Fig. 1(b). It was then propagated through the specimen using the multislice method [14]

$$\begin{aligned} \psi_{n+1} &= \mathcal{P}_{\Delta z} \otimes \big[\, t_n(\mathbf{r})\, \psi_n(\mathbf{r}) \,\big], \\ t_n &= \exp\big[\, i\,\sigma\, v_n(\mathbf{r}) \,\big], \end{aligned} \tag{2}$$

where v_n is the projected potential of slice n, σ the interaction constant and $\mathcal{P}_{\Delta z}$ the Fresnel propagator. The recorded quantity is $I(\mathbf{r}_p, \mathbf{k}) = |\mathcal{F}\psi_N|^2$, and this map from structure to diffraction data is the forward operator $\mathcal{S}$ that the reconstruction inverts. Calculations used abTEM [15] on GPU with Lobato potentials [16]. Thermal motion entered as a frozen-phonon average [17] over 16 displaced configurations, every species given the same root-mean-square displacement of 0.08 Å ($B = 0.17$ Å^2). The coherent reconstruction that carries the benchmark of Sec. III A contains no thermal motion at all; the frozen-phonon average applies only to the thermally averaged volumes used for the dose series. A finite electron budget was imposed as shot noise over $D = 10^4$–10^{10} e Å^{-2}.

C. Multislice ptychographic reconstruction

Ptychography recovers the object by iteration: propagate a trial object and probe to the detector, compare with the recorded intensity, correct, and repeat [18]. A single transmission function cannot represent a 70 Å specimen, so the object was carried as a stack of N_L

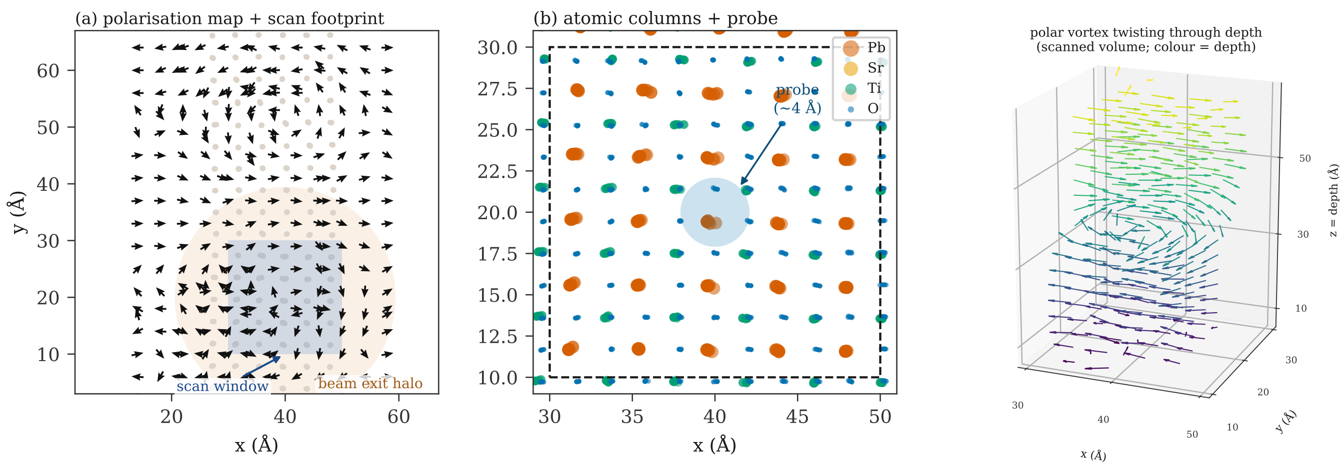

FIG. 1. The model system. (a) In-plane polarisation over the whole cell viewed down the beam, one arrow per atomic column, resolving the polar domains and vortices. Arrows are unit length and carry direction; colour gives the column-averaged magnitude, which is small where the polarisation reverses along the beam and the average cancels. The outermost ring of columns is omitted, the labyrinth being truncated at the box edge so that the oxygen cage there is asymmetric. The shaded square indicates the 20 Å scan window and the shaded disc the beam exit halo, the exit-wave footprint spanned by the recorded diffraction, which is why the full box was simulated. (b) Atomic columns in beam projection within the scan region, with the overfocused probe (≈ 4 Å diameter) overlaid; the lead columns split into polar dumbbells. (c) Polarisation of the scanned volume in three dimensions, with the local dipoles (arrows, coloured by depth) twisting into a vortex tube along the beam. This depth variation is integrated away by a projection.

slices with a propagation between each pair [4]; that stack is what makes depth recoverable. Slices and probe were solved with the LSQ-ML engine of Odstrčil *et al.* [19] within PtychoShelves [20], minimising the variance-stabilised amplitude metric [21] appropriate to the shot-noise model, using $N_L = 70$ slices for the coherent volume and 105 for the thermally averaged one. No regularisation was applied between layers, so the axial structure is set by the data and not by a smoothness prior that would bias the coordinate being measured. The probe is known exactly here, having been simulated, so it was held fixed as a single coherent mode; experimental data would instead need it solved as a mixed state [22, 23], because a real probe is never known that exactly. Parameters are collected in Table A1 of Appendix A.

D. Atom localisation

The reconstructed object was treated as a sparse superposition of identical single-atom responses,

$$\phi(\mathbf{r}) = \sum_{i=1}^{N} \beta_i \, K(\mathbf{r} - \mathbf{r}_i) + b(\mathbf{r}) + \varepsilon(\mathbf{r}), \qquad \beta_i \geq 0, \quad (3)$$

where ϕ is the per-layer median-subtracted phase, $(\beta_i, \mathbf{r}_i)$ the amplitude and three-dimensional position of atom i, and b a slowly varying background removed by a high-pass filter. Equation (3) is an assumption rather than a derived identity. The reconstruction is nonlinear, so treating its output as a sum of identical responses is an approximation, and one this work tests directly in Sec. III A before relying on it.

The kernel K was measured rather than parameterised. A sparse grid of one element at one depth was put through the byte-identical simulation and reconstruction pipeline, and one reconstructed blob taken as K. A grid rather than a lone atom is deliberate: a single atom leaves the unregularised reconstruction so underdetermined that it fills the volume with noise, whereas a 4 Å grid converges while still keeping every atom isolated, in-plane and in depth, at the $\approx$1 Å scale of the response. Measuring K costs one extra simulation and removes a modelling choice: an anisotropic Gaussian with a missing-cone tail, fitted as a stand-in, comes out too broad by a factor of two axially and four in-plane. Lead and titanium were both measured this way, and their shapes differ enough that the difference itself carries the species signal. Oxygen's own response is too close to the reconstruction noise to serve as a kernel, so it borrows the titanium shape (Appendix D). Detection and refinement run on the lead kernel, the species-resolved stage on each species' own, and the amplitude β carries Z. The species set is closed: every accepted atom is assigned one of the three elements known to be present, so the method places an atom rather than identifies it, and an unexpected element or an impurity would be forced into whichever of the three it resembled most.

Estimation proceeds in two stages. Matching pursuit on the dictionary $\{K(\cdot - \mathbf{r})\}$ [24] selects the support, stopping at a floor set by the local matched-filter noise so that it transfers between reconstructions of differing amplitude. Block-coordinate descent then refines each atom by Gauss–Newton steps on the local residual with all other atoms subtracted, so overlap is handled to first order. Candidates are kept on goodness of fit rather than amplitude, which preserves weak but well-formed responses. Column typing, the re-detection of empty lattice sites and every threshold are in Appendix B.

E. Uncertainty quantification

Uncertainty was built in two stages. The first computes a standard error from the fit alone, with no reference to the known structure. The second turns that number into an interval with a stated coverage.

The model error has two terms, $\sigma^2 = \sigma^2_{\text{stat}} + \sigma^2_{\text{sys}}$. The statistical term is the Cramér–Rao bound, the best precision any unbiased estimator could reach on this data. It was computed by assembling the Jacobian over all the atoms in a subvolume and solving once, rather than atom by atom as in the per-peak bound used to predict single-dopant precision [7]. On a dense sublattice the per-peak form assumes every neighbour is known exactly, and is optimistic by the variance inflation factor of the overlapping templates, which reaches 2.02 in depth at the 1.95 Å Ti–O separation. Solving for all of them at once recovers that factor with no tuned constant. The systematic term σ_{sys} is the position spread induced by kernel mismatch, measured by refitting one kernel's response with the other. Neither term uses the known structure, so both carry over to experimental data unchanged. That σ is a standard error, not yet an interval. To turn it into one, the atoms were split in two: half were used to measure how far the true error actually runs in units of σ, and that ratio was applied to the other half. This is split-conformal calibration [25], and it assumes only that the two halves are drawn alike, not that the error is Gaussian, which Sec. III B shows it is not. Measuring the ratio separately for each species, detection mode and depth band rather than once over all atoms is the Mondrian, or label-conditional, variant, and it is what makes coverage hold within each of those groups instead of only on average across them. The 95% interval is the default in released coordinates, and per-stratum coverage is exported to flag a calibration that fails to transfer. The derivation, the measured inflation factors and the coverage tables are given in Appendix C.

F. Polarisation from located atoms

Equation (1) was evaluated on the located atoms exactly as on the model, the six nearest located oxygens within 2.8 Å forming the cage. Titanium whose cage is incomplete in the reconstruction was excluded and reported separately rather than completed from the lattice. Uncertainty on $\boldsymbol{\delta}$ was propagated by Monte Carlo over 400 draws of the per-atom conformal half-widths of Sec. II E. Those half-widths need no ground truth, so neither does the error bar on $\boldsymbol{\delta}$.

G. First-principles EELS

Ptychography struggles with the along-beam component, so the second phase asks whether spectroscopy can recover it. When a core electron is ejected, the fine structure just above the edge, the energy-loss near-edge structure (ELNES), is shaped by the empty states it can be promoted into. Which of those states contribute is set by the direction of the momentum transfer $\mathbf{q}$: to a good approximation the edge measures the empty states of p character pointing along $\hat{\mathbf{q}}$ [26, 27]. For a near-axis collection $\hat{\mathbf{q}}$ is essentially the beam direction, so in a crystal with a polar axis the edge depends on how that axis is oriented relative to the beam. This is the signal the second phase is after, and two of its properties matter later. The selection is even in $\mathbf{q}$, so the edge reports the size of the along-beam polarisation but not its sign. And a real measurement averages over the convergence and collection apertures, with a weight that vanishes at the magic angle $\beta^\star \approx 4\theta_E$, where $\theta_E = E/(\gamma m_0 v^2)$ [28].

Spectra were computed with CASTEP [29] using the core-hole method [30] in a $2 \times 2 \times 2$ supercell, which dilutes the hole from its periodic images, and contracted by OptaDOS [31] into a core-loss spectrum with $\hat{\mathbf{q}}$ set explicitly. The quantity of interest is the ELNES dichroism

$$\Delta(E) = S(\mathbf{q} \,\|\, c, E) - S(\mathbf{q} \perp c, E), \tag{4}$$

which by reciprocity is the change in the measured edge when the polarisation rotates from along the beam to in-plane. The O K edge was taken as the primary observable. It is an established probe of ferroelectric off-centring in PTO/STO superlattices at the unit-cell scale [32]. Ti $L_{2,3}$ was used only to check anisotropy trends, since DFT of this kind does not reproduce the multiplet structure of that edge [33]. Each core-hole self-consistent field (SCF) calculation and its OptaDOS pass ran on the Warwick Blythe cluster. Both this calculation and the aperture model of Appendix F assume the electron scatters once, so they describe a specimen thin enough for plural scattering to be neglected; in a thick one it would dilute the effect.

III. RESULTS AND DISCUSSION

A. The measured response and localisation performance

The reconstructed lead response (Fig. 2(a)) has a full width at half maximum of 0.10 Å in-plane against 1.0 Å along the beam on the coherent reconstruction, and 2.0 Å on the thermally averaged one. Because that axial width is comparable to or larger than the 1.95 Å Ti–O spacing, neighbouring responses overlap. Equation (3) was

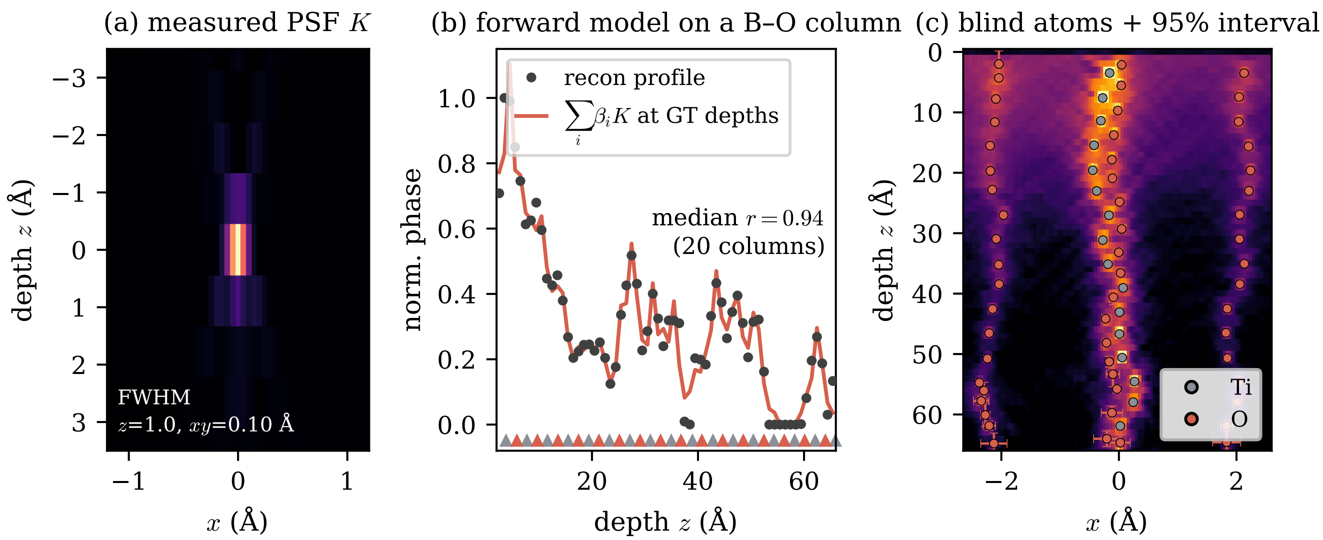

FIG. 2. Model-based localisation. (a) The measured lead point-spread function (PSF) K in a depth–x slice: tight in-plane, elongated along the beam. (b) Test of the forward model, Eq. (3). The measured depth profile of a mixed Ti–O column is compared with the superposition $\sum_i \beta_i K$ of that response at the known depths, matching at a median $r = 0.94$ over 20 columns; triangles mark the true Ti (grey) and O (red) depths. The apparent axial blur is overlap of a fixed response rather than lost information. (c) Blind atoms on a raw cross-section, coloured by assigned species, with calibrated 95% depth intervals. The intervals are strongly heteroscedastic and are visible only near the exit surface, where they reach ≈1.5 Å against ≈0.5 Å in the bulk.

	recall (%)			oxygen split (%)	
method	Pb	Ti	O	isolated	overlapped
3-D peaks, raw	93	88	71	91	26
RL deconv. + peaks	93	87	58	79	9
MEM deconv. + peaks	93	89	59	79	13
3-D model fit	88	89	89	92	82
unconstrained only	88	89	86	92	71
bulk band	95.6	96.4	95.5	—	—

TABLE I. Recovery of each sublattice, whole field unless stated, on the same reconstruction with the same measured kernel; the bulk band is $z = 10$–56 Å. Isolated oxygen occupies its own column, overlapped oxygen shares one with titanium 1.95 Å away; that split was scored over the whole field only, hence the dashes in the bulk row. Precision is 0.97 for the model fit against 0.84–0.99 for the baselines, and only the model fit returns species labels and per-atom uncertainties.

tested by fixing $\{\mathbf{r}_i\}$ at the known positions and solving for $\{\beta_i\}$ alone. A comb of measured kernels at the true depths reproduces the observed depth profile of a mixed B–O column at a median $r = 0.94$ over 20 columns (Fig. 2(b)), and at 0.92 when the whole two-dimensional patch is compared instead of the profile down its centre. The apparent axial blur is therefore kernel overlap rather than lost information.

Table I scores the method against the known structure, alongside the established alternatives, run on the same volume with the same measured kernel and each given the threshold that scored it best. Over the bulk ($z = 10$–56 Å, excluding the entrance and exit artefact bands) the model fit recovers 95.5% of oxygen, 96.4% of titanium and 95.6% of lead, with 1.1% of species labels incorrect, an in-plane RMS accuracy of 0.032 Å and a depth RMS of 0.37 Å. Figure 2(c) shows the located atoms on a raw cross-section, carrying the calibrated intervals of Sec. II E.

The oxygen sublattice was split into two populations by geometry. Oxygen in its own column, in-plane isolated, is found by every method here at 79–92%. Oxygen sharing a column with titanium 1.95 Å away is the difficult case. The methods separate here: 82% for the model fit against 26% for raw peak-picking, 13% for maximum-entropy deconvolution and 9% for Richardson–Lucy.

Neither deconvolution route behaved as expected. Deconvolution before detection does not help, and Richardson–Lucy makes the oxygen worse, both routines losing oxygen relative to peak-picking the raw volume at 58–59% against 71%. They redistribute dynamic range towards the strongly scattering species, so weak oxygen falls below any relative floor. Deconvolution only redistributes what was recorded, and a single orientation never records the axial frequencies that would separate an oxygen from the titanium stacked with it. Maximum entropy is reported as the stronger of the two routines [12], but no advantage over Richardson–Lucy appeared on this dense lattice, so the limitation looks structural rather than a matter of tuning. On noiseless data a good peak-picker is also competitive on the heavy atoms, since a raw local maximum is already a clean estimator of an isolated bright column, so Table I understates the margin rather than flattering it. Whether the margin widens with noise is not settled here, for the reason given in Sec. III C. A peak-picker cannot, at any dose, provide a species label or a calibrated error bar.

B. Calibrated uncertainty

Figure 3(a) quantifies why a per-peak bound is inadequate here. The variance inflation of the joint fit is negligible at the 3.9 Å heavy-atom spacing but reaches 2.02 in depth at the 1.95 Å Ti–O separation, so a per-peak bound understates σ_z by $\approx 1.4\times$ for every oxygen sitting directly above or below a titanium, and most atoms in the field have such a neighbour (Appendix C). Solving for all the atoms together, rather than one at a time, restores that inflation for each atom, with no constant left to tune.

Taking that σ at face value, as if the error were Gaussian, fails badly on near-noiseless simulation: an interval quoted at 68% actually contains only $\approx 8\%$ of atoms in depth (Fig. 3(b)). The fit residual is almost zero, so σ is almost zero, while the real error is set by systematics the residual cannot see. The calibration supplies that missing scale from the data. It has to be tested on atoms it has not seen, since the ones used to set the ratio are fitted by construction: calibrating on half the matched atoms and measuring on the other half, averaged over twenty random splits, gives 69/68/68% in (x, y, z) against a 68% target and 96/95/95% against 95% (Fig. 3(b)). Measuring the ratio per stratum matters, because the strata genuinely differ: in depth it runs from 6.7 to 14.4, a factor of 2.2, so one pooled number would leave some strata with intervals too wide and others too narrow. Pooling drops the worst stratum to 84% against its 95% target, while the per-stratum ratio brings every one back to 95–100% (Fig. 3(c)).

The decisive test is whether an uncertainty estimate survives a change of dataset. The identical code, with no tuned constants, was applied to an independent reconstruction of the same material at a different in-plane sampling (0.10 against 0.15 Å) and a different slice discretisation (105 against 70 layers), and carrying thermal diffuse scattering from sixteen frozen-phonon configurations and a finite electron dose where the coherent volume had neither. Its phase amplitude is roughly half as large in consequence. Re-calibrating on that volume's own atoms restores coverage to target, as it must, since the quantile is fitted there. What matters is a calibration carried across unchanged: the hand-tuned per-species error floors used in an earlier version of this work, calibrated to 68% coverage on the first volume, transferred to only 38/47/56% on the second. The ratios on that volume run from 10.9 to 23.3, wider still than on the first, which is exactly the spread a single fitted constant cannot represent.

These intervals are limited in scope. Each is conditional on the atom having been correctly detected and typed, so an atom that is not really there, or one given the wrong element, is a separate kind of failure and is reported through precision (0.97) and confusion (1.1%) instead. Conformal calibration also assumes exchangeability, which is why the workflow re-calibrates per volume.

dose ($\mathrm{e\,Å^{-2}}$)	found	precision	Pb	Ti	O	confusion
10^{10}	1575	0.85	89 %	74 %	60 %	12.4 %
10^{8}	1593	0.88	89 %	77 %	66 %	11.3 %
10^{6}	521	0.96	76 %	26 %	5 %	37.8 %
10^{4}	0	—	—	—	—	—

TABLE II. Dose series on an independent reconstruction (105 layers, 0.10 Å sampling), run with the same configuration as the coherent reference. Recalls are over the $z = 10$–58 Å bulk band of the 105-layer geometry. The 10^6 row illustrates the uninformativeness of precision, which reads 0.96 while oxygen recall is 5%. At 10^4 the method returns no atoms at all rather than reporting noise.

C. Behaviour at finite dose

Table II applies the same configuration over a dose series on the independent reconstruction. Performance is maintained to $10^8\,\mathrm{e\,Å^{-2}}$ and collapses below 10^6. The residual gap to the coherent reference (Ti 74% on the 105-layer bulk band against 96% on the 70-layer one) is dominated by the reconstruction and the kernel it yields rather than by the localisation constants: a 2.0 Å axial kernel against 1.0 Å, and atomic planes that stand out less sharply along the beam. The latter is measured by matching each column's depth profile against a comb of peaks at the known atom depths, which correlates at 0.44 on the dosed volume against 0.63 on the coherent one. A coherent kernel for that volume is the single most likely improvement, since every kernel available for it is thermally averaged.

Two lessons transfer beyond this dataset. Precision is not a safety indicator: at $10^6\,\mathrm{e\,Å^{-2}}$ it reads 0.96 while titanium recall is 26%, oxygen 5% and 38% of labels are wrong, because it measures only the surviving bright lead atoms, which are correctly placed. Species confusion and σ-coverage are the diagnostics that respond, and both are printed as health warnings. And abstention is worth more than a number: at 10^4 the model fit returns nothing, whereas maximum-entropy deconvolution returns 3803 detections at a precision of 0.34. The single-kernel qualification, the depth-registration coupling and the baseline comparison over this series are set out in Appendix D.

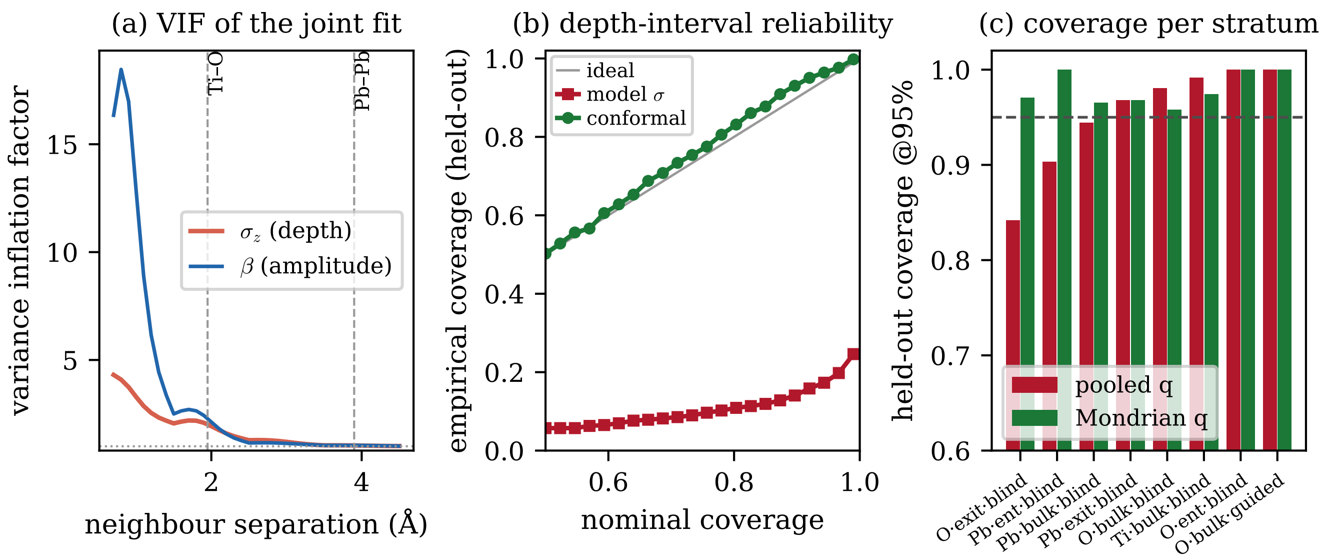

FIG. 3. Calibrated uncertainty, from a 50/50 split-conformal calibration/test split. (a) Variance inflation factor (VIF) of the joint fit against neighbour separation: near unity at the 3.9 Å heavy-atom spacing, ≈2 in depth at the 1.95 Å Ti–O spacing. (b) Held-out reliability of the depth interval: the raw model standard error treated as Gaussian (red) severely under-covers, whereas the conformal interval (blue) tracks the ideal diagonal. (c) Held-out coverage at 95% for the eight strata where a pooled quantile (red squares) departs most from target; the label-conditional quantile (blue circles) restores each.

D. Polarisation with error bars

Equation (1) needs a full oxygen cage, and only 70% of the titanium present acquire one from blindly located oxygens. That leaves 171 of the 247 titanium in the bulk band to be scored in Fig. 4. The column-averaged map of Fig. 4(a) tracks the ground truth beside it, and both are small over much of the field: the mean in-plane direction turns from $-170°$ over $z = 10$–30 Å to $-5°$ over $z = 40$–70 Å, so a column average through that reversal cancels. This is the projection loss the three-dimensional map exists to avoid.

The in-plane off-centring is recovered essentially exactly (Fig. 4(b)). Against a true magnitude of 0.274 Å the median vector error is 0.007 Å, and the direction, from which a vortex map is built, is good to 0.9°. The two components are not alike, since the signal lies mostly along x: its true spread is 0.272 Å and it correlates with truth at 0.96, against 0.078 Å and 0.80 for δ_y, which also carries a +0.014 Å bias. Even so δ_y is measured, its spread exceeding the propagated $\sigma_y = 0.009$ Å ninefold, and that is the comparison to hold against the along-beam component below.

Five per cent of titanium have a cage of the wrong composition, the open circles of Fig. 4(b), since a missing or misassigned oxygen displaces the centroid, and for these the in-plane error rises to a median of 0.45 Å, comparable to the signal itself. The propagated interval covers 99% of the well-caged titanium but only one of the eight in the tail. The interval is conditional on correct detection, so a gross error in cage composition is a detection failure, to be reported through precision, confusion and cage completeness rather than absorbed into a positional error bar. The 30% of true titanium that acquire no complete cage should likewise be shown as absent rather than interpolated.

The along-beam component behaves in the opposite way (Fig. 4(c)), and this is the more consequential result. Its true spread across the scanned volume is small, $\sigma = 0.074$ Å, since in this zone axis the vortex polarisation is predominantly in-plane, whereas the propagated uncertainty on δ_z is 0.237 Å, more than three times the entire signal, and the correlation with truth is correspondingly poor at $r = 0.42$ with a median error of 0.122 Å. The along-beam component is therefore not measured by this experiment. The calibrated uncertainty establishes this without ground truth, since comparing the propagated σ_z with the spread of the recovered δ_z requires no reference structure: the along-beam map is flagged as uninformative while the in-plane map is good to 0.01 Å. Propagated intervals cover 95/93/99% of true errors in (x, y, z) at a 95% target, so neither the confidence in-plane nor the pessimism along the beam is overstated.

Depth localisation of individual atoms is itself good, at 0.37 Å RMS and well inside the ≈ 2 Å depth resolution, as single-molecule localisation improves on the diffraction limit [34]. A polarisation is, however, a difference of two such coordinates, and differencing two 0.3 Å depth errors cannot resolve a 0.07 Å displacement.

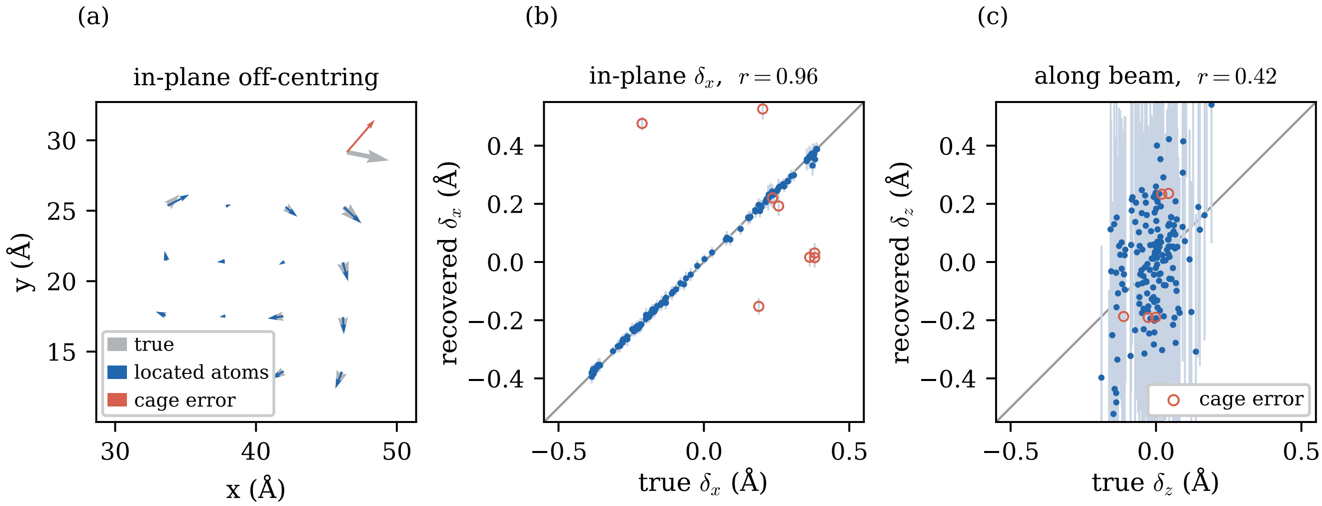

FIG. 4. Local polarisation from the located atoms. (a) In-plane Ti–O_6 off-centring averaged per atomic column, drawn to scale: recovered (blue) over ground truth (grey, drawn thicker so the two overlie where they agree). The 171 scored titanium fall into nineteen columns, of which the seventeen holding at least three are drawn; a mean over one or two atoms is not a column mean. Their median column-averaged magnitudes agree at 0.105 against 0.106 Å. Both fields are short because the polarisation reverses with depth, so the column average largely cancels. The single column drawn in red is the one the wrong-cage tail below dominates, and is also the only one whose direction visibly disagrees; every other column matches truth to within 0.07 Å. (b) Recovered against true in-plane off-centring, component δ_x, with propagated 95% intervals; open red circles are the 5% of titanium whose oxygen cage has the wrong composition. (c) The same along the beam, where the propagated uncertainty exceeds the entire spread of the component itself.

E. Can EELS see the missing component?

The second phase examines whether the along-beam polarisation leaves a spectroscopic signature. Each step of the calculation was validated before the next was run. CASTEP/PBE places the polar structure 203 meV per formula unit below the equally strained centrosymmetric reference, the core-hole recipe reproduces the rutile TiO_2 O K edge, and the cubic cell returns a dichroism of 0.02% at its isotropic Ti site, so any anisotropy found later is physical rather than numerical (Appendix F).

The two limiting orientations of the polar cell are those of Fig. 5(a). With the polar axis along the beam the apical-oxygen O K edge is strongly dichroic, at $\max|\Delta|/\max S = 78.2\%$ and $\int|\Delta|/\int S = 62.8\%$ (Fig. 5(b)). The origin is a σ/π anisotropy: with $\mathbf{q} \perp c$ the edge shows a sharp π^* peak ≈ 8.5 eV above onset from O $2p_{x,y}$, whereas with $\mathbf{q} \parallel c$ it shows σ^* features at ≈ 11 and 18.5 eV from the O $2p_z$ states directed along the Ti–O–Ti chain. The edge of the cell as a whole is the multiplicity-weighted sum of one apical and two equatorial oxygen, and the equatorial site behaves differently: its own chain lies in-plane, and it is only 22% dichroic. Carrying twice the multiplicity, it brings the weighted full O K edge to 35.8% rather than 78%, with no free parameter left in the combination (Appendix F). The rotational-invariance test passes: mapped into the crystal frame the $c \parallel z$ and $c \parallel x$ cells agree to 0.02–0.03%, so the anisotropy is physical rather than an artefact of the momentum-transfer machinery.

A large orientation dichroism does not by itself constitute a polarisation measurement, since the tetragonal backbone is anisotropic even without off-centring, as the grey reference curves of Fig. 5(b) show. Holding the lattice parameters fixed and varying only the internal displacement s separates the two contributions. At $s = 0$, corresponding to zero polarisation with full tetragonal strain, the dichroism is already 70.1%, rising to 78.2% at $s = 1$. Approximately 70 of the 78 points are the strain-induced backbone anisotropy, and the off-centring supplies the remaining 8 together with a monotonic 18.7% change in the along-beam-probed O K spectrum, 4.3% of it by $s = 0.5$. This is the first-principles counterpart of the empirical observation that the O K edge responds to titanium off-centring [35], and it makes the measurement a polarisation probe rather than only a strain probe.

The intrinsic dichroism must then survive the acquisition geometry, and the textbook treatment of that is misleading here. It assumes parallel illumination, the only geometry in which a magic angle is defined, and predicts that collecting out to the $\beta \approx 100$ mrad a hollow detector allows would average the anisotropy away. A focused probe is not that geometry: the momentum transfer is a vector difference, so the convergence semi-angle spreads $\mathbf{q}$ however small the spectrometer aperture is. Averaging over both cones (Appendix F) shows that closing the collection aperture is not by itself a remedy, and that beyond the magic angle the anisotropy inverts and saturates

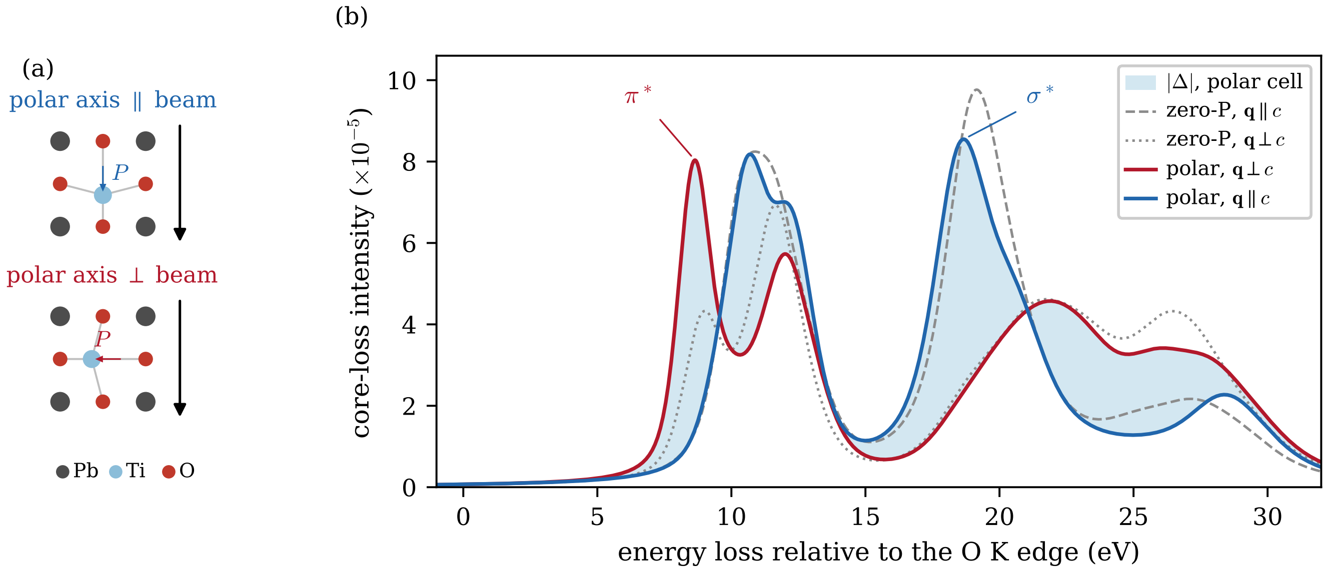

FIG. 5. Orientation dichroism of the $PbTiO_3$ apical-oxygen O K edge, from core-hole CASTEP and OptaDOS. (a) The polar cell in the two limiting orientations, with the beam vertical and the arrow on the titanium marking its off-centring. (b) The corresponding edges: coloured curves are the polar cell probed across ($\mathbf{q} \perp c$) and along ($\mathbf{q} \parallel c$) the polar axis, and the shaded band between them is the dichroism $|\Delta|$ of Eq. (4); grey curves are the same two orientations for the zero-polarisation, equally strained reference, which isolates the backbone contribution. Energy is referred to the O K edge, and all four spectra are on the single OptaDOS normalisation that makes $\max|\Delta|/\max S$ meaningful.

near a third of its intrinsic value rather than vanishing. At the $\alpha = 100\,\mathrm{mrad}$ convergence that atomic-resolution imaging requires the surviving fraction is flat in β, so a domain with its polar axis along the beam differs from one with the axis in-plane by 12.1% in the collected O K edge whatever the spectrometer accepts. The aperture is therefore not a design variable: a hollow detector that gives the spectrometer the whole bright-field disc loses nothing a smaller one would recover.

Two further limits are independent of the aperture. The dipole ELNES is even in $\mathbf{q}$, so it measures the magnitude of the along-beam polarisation but not its sign. A dynamical multislice core-loss simulation of up- and down-polarised domains gives O K spectra identical to within 1.0%. That simulation injects the computed dipole $\sigma(E)$, so it cannot test the parity itself; what it shows is that channelling through the specimen does not break the degeneracy either. In the labyrinth itself, viewed along this zone axis, the polarisation is in any case mostly in-plane: over 2560 polar titanium the median angle from the beam is 82° (interquartile range 73–87°), an along-beam fraction of only ≈ 0.14.

The computed dichroism is not an unrealistic upper bound. Repeating the calculation on a cell built at the labyrinth's median 82° tilt gives a dichroism of 76.3%, against 78.2% for the fully aligned case. What changes is not the magnitude but the character: the σ^* feature at 18.7 eV and the π^* at 8.6 eV appear at the same two energies with their assignments exchanged when the polar axis moves from along the beam to in-plane. The edge reports the orientation of the polar axis relative to the beam, the sense of the asymmetry identifying which way it lies, rather than the magnitude of the along-beam component. A measurement on the labyrinth would therefore return a large dichroism reporting an in-plane axis, not an absence of signal.

IV. CONCLUSIONS AND FURTHER WORK

Fitting a reconstructed multislice-ptychographic volume as a superposition of a measured single-atom response recovers the oxygen sublattice of a PTO/STO superlattice where the established deconvolve-then-detect workflow does not, at 95.5% bulk recall over the same band against 68% for deconvolve-then-detect and 77% for peak-picking the raw volume. The whole of that margin lies in the axially overlapped oxygen (82% against 9–26%). This does not appear to be a tuning artefact, since the limitation of deconvolution is the axial spatial frequencies that a single-orientation measurement never records. The method also returns species labels (1.1% confusion) and calibrated per-atom uncertainty, neither of which a peak detector provides.

The uncertainty machinery is arguably the more transferable contribution. A joint Cramér–Rao bound recovers the twofold variance inflation that a per-peak bound misses wherever responses overlap, and Mondrian conformal calibration converts it into intervals holding their nominal coverage per stratum. These contain no hand-tuned constant and transfer unchanged to an independent reconstruction, on which hand-tuned error floors give only 38–56% coverage against a 68% target. Coverage should therefore be reported rather than a fitted σ: the same pipeline that reads σ-calibrated at one dose reads 38% species confusion at another, while its precision still reads 0.96.

The in-plane polarisation map is quantitative, with a median error of 0.007 Å in magnitude and 0.9° in direction against a 0.27 Å signal. The along-beam component, by contrast, is not measured at all, because differencing two 0.3 Å depth coordinates cannot resolve a 0.07 Å displacement. The calibrated uncertainty establishes this without reference to the known structure. First-principles core-hole DFT indicates that the missing component is spectroscopically accessible in principle: the multiplicity-weighted O K edge is 35.8% dichroic to the polar-axis orientation, of which a monotonic 19% change is the signature of the displacement rather than of the tetragonal backbone. Two limits remain. The dipole term is even in $\mathbf{q}$ and so forbids recovery of the sign, and the intrinsic anisotropy is largely averaged away in any geometry a focused probe can provide.

Fitting oxygen with the titanium kernel is weakest where it matters most, since the larger Debye–Waller factor of oxygen broadens its response under thermal scattering, and in-situ vacancy-difference kernels would supply an in-crystal oxygen matched filter. The 5% cage-composition tail of Sec. III D is a detection rather than a localisation problem and needs a calibrated cage-completeness criterion, while learned denoising before fitting [36] is a natural route below $10^6\,\mathrm{e}\,\text{Å}^{-2}$. The clearest instrumental conclusion is that a hollow detector is the right geometry, though not for the obvious reason: the convergence and not the collection aperture sets the averaging, so nothing is lost by giving the spectrometer the central disc and the imaging the outer annulus. The limit on such an experiment in this material is not the detector but the specimen: the polar axes of the labyrinth are clustered about the in-plane direction, so the orientation contrast between neighbouring domains is of order 1% rather than the 12% available between the limiting orientations, and a zone axis presenting a wider spread of axes would serve far better. Whether the two can be fused into a single three-dimensional vector polarisation measurement remains open, and combining ptychography and spectroscopy on the same oxygen sublattice [9] suggests one route. The pipeline itself assumes only that the reconstructed volume is a superposition of a measurable single-atom response, and that a calibration set with known positions is available and exchangeable with the target. Neither condition is peculiar to a ferroelectric superlattice.

Appendix A: Simulation and reconstruction parameters

Table A1 collects the parameters of the forward simulation, the reconstruction and the first-principles calculations. Reproducing the reconstruction from the simulated data also depends on the geometry conventions recorded here. The object pixel is set by the diffraction angle rather than by the scan step. Each binned pixel subtends 1.125 mrad, so the 356^2 detector reaches ± 200 mrad, the 100 mrad bright-field disc has a radius of 89 pixels and the object pixel is $\mathrm{d}x = 0.0492$ Å. The detector requires a single transpose relative to the PtychoShelves convention. A two-stage schedule, a downsampled 178-pixel pre-solve followed by a full-resolution 356-pixel refinement, accelerates convergence of the nonconvex objective.

Appendix B: Localisation algorithm

The five stages below use the reconstructed phase volume, the measured kernel and the crystallography of the material. No ground-truth coordinate enters at any point.

1. Preprocessing: Each layer had its median subtracted, so that vacuum mapped to zero, and a heavily Gaussian-blurred copy of the layer was removed to suppress the slowly varying channelling haze that otherwise biases weak oxygen amplitudes. The result was clipped at zero and the entrance and exit artefact layers ($z < 2$ Å, $z > 66$ Å) trimmed.

2. Column seeding: In-plane the kernel is only two pixels wide, so local maxima of the depth-averaged image over a 1.4 Å non-maximum window gave $\approx$ 98 column seeds across the field.

3. Support selection: Around each seed a fixed tube was taken, ± 0.5 Å in-plane and spanning the full interior depth, and matching pursuit run within it with the unit-norm kernel: the residual was correlated with K, the global maximum accepted as an atom, βK subtracted, a neighbourhood of ± 1.5 layers by ± 0.2 Å excluded, and the process repeated. Iteration stopped at $\max(K \star R) < k\,\sigma_{\mathrm{MF}}$, where σ_{MF} is a median-absolute-deviation estimate of the matched-filter noise taken over the sub-median voxels of the same tube and $k = 2$. Because each atom carries its own (x, y, z), the ferroelectric lean of a column costs nothing.

4. Joint refinement: Two sweeps of Gauss–Newton refinement follow. For atom i the models of all other atoms were subtracted and (β, x, y, z) fitted on the local patch by linearising K, with $\partial K/\partial x$ and its counterparts obtained by central differences on the measured kernel, giving sub-voxel positions. A candidate was rejected if the normalised correlation between its patch and βK fell below 0.5. The criterion tests shape rather than amplitude, since an amplitude cut cannot separate faint oxygen from a noise peak of the same height.

5. Species assignment and lattice-constrained detection: Columns self-classify by k-means on their own per-column amplitude statistics, whose 75th percentiles are well separated in this material at ≈ 4.1, ≈ 1.5 and ≈ 0.66 for lead, mixed B–O and pure-oxygen columns. On B–O columns titanium and oxygen alternate every ≈ 1.95 Å, and species were assigned by per-column amplitude with a dead zone, ambiguous atoms resolved by local parity, that is by $z_i - z_j$ modulo the fitted period to confident neighbours. Parity is used rather than a global comb phase, which accumulates period error towards the column ends and inverts the labels there. Expected sites were then enumerated by extending $\pm kp$ from each found atom of the same species and $\pm p/2$ from each titanium for oxygen; where a site is empty, meaning further than 0.8 Å from any same-species atom, a matched-filter search runs on the residual within gates of ± 0.7 Å in depth and ± 0.35 Å in-plane, followed by refinement. Such a detection was accepted only if its fit quality cleared a reduced bar of 0.35 and its amplitude lay between 0.45 and 1.6 times the column's same-species median. The reduced bar was permitted for oxygen alone; applied to lead and titanium it produced mislocated atoms with overconfident errors and was abandoned. A post-fit occupancy guard rejects any constrained detection landing within 1.2 Å of an atom already accepted in the same tube, since no two atoms in the reference structure are closer than 1.67 Å. All such atoms carry a flag through to the exported coordinates, so the lattice prior is always visible in the output.

Appendix C: Uncertainty quantification

1. Joint versus per-peak Cramér–Rao bound

For the stacked parameter vector $\theta = (\theta_1, \ldots, \theta_N)$, with $\theta_i = (\beta_i, \mathbf{r}_i)$, the Fisher information of Eq. (3) under homoscedastic residuals is $\mathbf{F} = \mathbf{J}^\top \mathbf{J}/s^2$, and the attainable variance of an unbiased estimate of θ_i is the corresponding diagonal block of $\mathbf{F}^{-1}$ rather than the inverse of the diagonal block $\mathbf{F}_{ii}$. The two coincide only for an orthogonal design; in general $[\mathbf{F}^{-1}]_{ii} \geq [\mathbf{F}_{ii}]^{-1}$, and the ratio is the variance inflation factor (VIF) of the overlapping templates. Evaluating $[\mathbf{F}_{ii}]^{-1}$, in which each atom is fitted with its neighbours held fixed, is the per-peak bound used to predict single-column precision [13] and single-dopant depth [7]; on a dense sublattice it is the bound obtained with all neighbours known exactly. Table C1 gives the inflation measured for our kernel. Since

Stage	Parameter	Value
Sample	composition	$PbTiO_3/SrTiO_3$, 19 440 atoms
	supercell	$70.0 \times 70.0 \times 47.9$ Å
Probe	energy / α	300 keV / 100 mrad
	defocus C_{10}	-20 Å (20 Å overfocus)
	aberrations	none ($C_{30} = C_{12} = 0$)
	coherence	fully coherent
Scan	window / step	20 Å / 0.10–0.15 Å
Sim.	slice thickness	2.0 Å (0.5 Å fine)
	frozen phonons	16 configs, $\sigma = 0.08$ Å
	detector	± 200 mrad, 356^2 px, 1.125 mrad/px
	dose	10^4–10^{10} e Å^{-2} (Poisson)

Stage	Parameter	Value
Recon	engine	PtychoShelves LSQ-ML
	layers N_L	70 / 105 ($\Delta z = 1.0$ / 0.67 Å)
	regularisation	inter-layer off; $\beta_{\mathrm{LSQ}} = 0.1$
	probe	1 mode, held fixed
DFT	code / functional	CASTEP [29] / PBE
	plane-wave cutoff	800 eV (swept 600–900)
	k-grid	$4 \times 4 \times 4$ MP (SCF and spectral)
	empty bands	`nextra_bands` = 100
	core hole	full ($1s^1/2p^5$), charge +1
	supercell	$2 \times 2 \times 2$ (40 atoms)
ELNES	code / task	OptaDOS [31], core-loss
	$\hat{\mathbf{q}}$ directions	$\langle 001\rangle$, $\langle 100\rangle$, $\langle 010\rangle$
	DOS spacing	0.01 eV, adaptive smearing 0.2
	broadening	Gauss 1.0 / Lorentz 0.5 eV, scale 0.1

TABLE A1. Simulation, reconstruction and first-principles parameters. The probe carries only defocus, so the simulated instrument is an ideal aberration-corrected one, and the illumination is fully coherent with no source size or energy spread imposed. The only incoherence in the data is therefore thermal diffuse scattering from the frozen-phonon average, applied to the thermally averaged volumes alone, and Poisson counting noise. Broadening is the OptaDOS lifetime-and-instrument model: a Gaussian instrument width, a Lorentzian core-hole lifetime width, and an energy-dependent scaling of the latter.

sep.	pair	VIF(β)	VIF(z)	σ_z low by
3.90 Å	Pb–Pb, Ti–Ti	1.04	1.04	$\times 1.02$
1.95 Å	Ti–O (apical)	2.28	2.02	$\times 1.42$
0.99 Å	near-degenerate	13.3	3.33	$\times 1.83$

TABLE C1. Variance inflation of the joint fit relative to the per-peak bound, measured on the kernel used here. Every oxygen stacked with a titanium sits at the 1.95 Å separation.

85% of atoms in the field have a neighbour within 2.5 Å, the per-peak form is optimistic almost everywhere. We therefore assemble $\mathbf{J}$ over all atoms in a subvolume and invert $\mathbf{F}$ once, a $4N \times 4N$ system with $N \leq 45$ and of negligible cost, taking each atom's marginal variance.

The systematic term σ_{sys} is obtained by generating a synthetic response with one measured kernel, refitting it with the other measured kernel, and taking the resulting position spread. Being computable without ground truth, it transfers to experimental data. On the coherent reconstruction, where the lead and titanium kernels are nearly identical, it is small ($\sigma_z = 0.02$ Å); it grows on datasets whose kernels genuinely differ. It cannot be formed at all where only one kernel is available, as on the dosed volumes, and is set to zero there.

2. Conformal calibration and measured coverage

Nonconformity scores $s = |\Delta|/\sigma$ were formed per axis and the empirical $(1-\alpha)$ quantile taken with the finite-sample correction, so the reported half-width is $q_{1-\alpha}\sigma$. Coverage is guaranteed only marginally, and the error is strongly heteroscedastic across the lattice, so calibration was performed per stratum, with strata of species $\times$ detection mode $\times$ depth band. Table C2 gives the measured held-out coverage on the coherent reconstruction and on the independent dosed volume. On the latter the conformal step is re-run on that volume's own matched atoms, which is the intended workflow, since applying one volume's quantile table to another assumes an exchangeability that fails when registration quality differs. The per-stratum quantiles $q_{68}(z)$ span 6.7–14.4 on the coherent volume and 10.9–23.3 on the dosed one.

volume	target	conformal	floors
coherent, held out	68%	69/68/68%	—
coherent, held out	95%	96/95/95%	—
dosed, re-calibrated	68%	at target	38/47/56%
dosed, re-calibrated	95%	at target	—

TABLE C2. Coverage of the calibrated intervals. The coherent rows are held out: the quantile is fitted on half the matched atoms and coverage measured on the other half, averaged over twenty random partitions (spread ± 2 and ± 1 percentage points at the two levels). The dosed rows are the transfer case, in which the quantile is re-fitted on that volume's own atoms, so its coverage sits at target by construction; the informative quantity there is the final column, the hand-tuned per-species error floors used in an earlier version of this work, calibrated to 68% on the coherent volume and applied unchanged to the dosed one.

Exported per atom are the model σ, the 95% conformal half-width taken as the default, and a within-column amplitude posterior p_{species} for the assigned label. The last is a relative confidence rather than a calibrated probability, and it is undefined in the useful sense on single-species columns, but atoms with $p_{\mathrm{species}} < 0.9$ are enriched for mislabelling by roughly $25\times$ over the 1.1% base rate.

Appendix D: Kernel provenance, the dose series and noise robustness

On the coherent reconstruction both the lead and the titanium kernels survive that procedure cleanly, with an axial full width at half maximum of 1.0 and 2.0 Å respectively and a three-dimensional correlation of 0.871. Oxygen does not. Its isolated response is about 1% of lead's and sits at the level of the reconstruction noise, so the extracted kernel is noise-dominated and its peak lands 1.15 Å from the atom. Oxygen is therefore fitted with the titanium kernel; a parameterised oxygen response is the obvious alternative and is left as further work. On the thermally averaged, dosed geometry the titanium and oxygen kernels have signal-to-noise ratios of 2.4 and 2.7, with their interior maxima displaced to the volume corner, and are unusable; the dose series of Sec. III C therefore runs on the lead kernel alone, which is why σ_{sys} vanishes on those volumes. A kernel derived instead from lead blobs within the crystal is measurably worse, its axial width inflated to 4.0 Å by unresolved neighbours, and within that comparison it reduces lead recall from 85% to 65%. Giving oxygen the titanium shape is safe on these volumes, which use one displacement amplitude for every species, but it is the assumption that would weaken first on experimental data, where the Debye–Waller factor of oxygen is the larger ($B = 0.80$ against $0.45\,\text{Å}^2$). One mismatch is real: the dose-series kernel grids were simulated with per-species amplitudes, so the lead kernel carries $\sigma_{\mathrm{Pb}} = 0.185\,\text{Å}$ against the 0.08 Å of the data, making it slightly the broader of the two and the dose recalls pessimistic rather than flattered.

Three qualifications attach to the dose series of Sec. III C. First, a single kernel, measured on the 10^{10} volume, was used at every dose: the response is a property of the imaging system rather than of the electron budget, and measuring it on a noisier volume would only add noise to the forward model. The series therefore tests whether the localisation constants carry across to a harder reconstruction, not whether the reconstructed response itself changes with dose, which it may, since the reconstruction is nonlinear. For the same reason σ_{sys} is zero on these volumes, leaving their model σ purely statistical before calibration. Second, species assignment is coupled to depth registration, since a 1 Å depth error on a lattice whose Ti and O alternate every 1.95 Å swaps labels wholesale: correcting the offset from 1.26 to 0.29 Å raised titanium recall from 53% to 74% and cut confusion from 19.0% to 13.7% at $10^{10}\,\mathrm{e}\,\text{Å}^{-2}$, with the occupancy guard of Appendix B taking it to the 12.4% of Table II. Sub-ångström depth accuracy is a prerequisite for correct species assignment.

Third, the baselines of Sec. III A were run over the same series, in the expectation that the oxygen margin would widen once noise was present. It does not, but the comparison is uninformative rather than adverse: peak detection is capped at a fixed number of maxima and reaches that cap at every dose, so its recall is set by the cap and not by the data. At 10^8 it buys the same true detections as the model fit, 1407 against 1404, with 1093 false ones against 189. A fair test would need the baselines re-thresholded to a matched count, so no claim is made from this series.

injected σ	found	precision	Pb	Ti	O	O overlapped
0	1834	0.97	88 %	89 %	89 %	82 %
0.01	1793	0.97	88 %	91 %	86 %	74 %
0.02	1421	0.98	87 %	90 %	57 %	37 %
0.04	795	0.96	87 %	86 %	5 %	18 %
0.08	408	0.95	87 %	0 %	0 %	0 %

TABLE D1. Injected-noise sweep on the coherent volume, all other settings fixed. Recalls are over the full depth rather than the bulk band. Species confusion rises from 1.1% at $\sigma = 0$ to 3.7% at 0.02, 58.7% at 0.04 and 74.6% at 0.08. The final column is the axially overlapped oxygen of Sec. III A, the population on which the method's advantage rests, and it is also the population the noise removes first.

Table D1 reports a controlled noise sweep in which Gaussian noise is injected into the reconstructed volume with geometry, kernel and calibration held fixed. The intrinsic vacuum noise floor of the volume is $\sigma \approx 0.023$ and the oxygen peak amplitude ≈ 0.06, so the collapse of oxygen recall between $\sigma = 0.01$ and $\sigma = 0.02$ corresponds to the oxygen signal falling below the noise. It is not a thresholding artefact and no choice of detection floor removes it. The sweep reproduces the diagnostic failure of Sec. III C on a second axis: at $\sigma = 0.08$ the recall of both titanium and oxygen is zero and three in four species labels are wrong, yet precision still reads 0.95. The failure is at least conservative, the lattice-consistency amplitude gate causing constrained oxygen detections to abstain rather than fabricate.

Appendix E: Reproducibility protocol

All code, the input structure and the per-run parameters are held in the project repository. The pipeline is scripted end to end, from simulation (`sim/simulate_4dstem.py`) and reconstruction (PtychoShelves `GPU_MS`) through localisation (`atomfind/run_atomfind.py`) and polarisation (`atomfind/polarisation.py`) to the DFT staircase (`eels/build_cells.py`, `run_coreloss.slurm`, `analyze_elnes.py`), with a unit-test suite guarding structure generation and parsing.

1. The nominated result

The result offered for independent reproduction is the localisation-with-uncertainty output of Sec. III D. The localisation code, its inputs and its protocol are together in a single self-contained repository,[1] which carries four inputs: the reconstructed phase volume as a $70 \times 404 \times 404$ array, the measured lead and titanium kernels, and the reference structure pre-transformed into the beam frame. Storing the reconstruction as its phase rather than as the complex object is exact, since the modulus is unused downstream and the phase of a single-precision complex array is itself single-precision, and it halves the size, which is what allows the data to be versioned with the code rather than distributed separately. The simulation and reconstruction that generate the volume require GPU resources well beyond a few hours and are therefore supplied pre-computed; everything downstream of them is reproduced from scratch. The whole procedure is

```
git clone <repository> && cd atomfind
python3 -m venv venv && . venv/bin/activate
pip install -r requirements.txt
python atomfind/test_atomfind.py
python atomfind/run_atomfind.py \
       --preset NL70_coherent --out ./out
python atomfind/polarisation.py --out ./out
```

executed from the repository root. The third command is a suite of seventeen tests that run on synthetic data in about a second and require none of the inputs, so that a disagreement with the values below can be told apart from a broken environment before the full run is attempted. The run requires a few minutes on a single core and no GPU, and depends only on NumPy, SciPy, h5py, Matplotlib and ASE. No path is hard-coded: the data files are located by name, through an environment variable, a directory adjacent to the package, or the `--data-dir` option.

These instructions were executed rather than merely written. The package was rebuilt from the repository, unpacked into a bare virtual environment holding no scientific software beyond those five libraries, and run end to end. It reproduced every quantity below, with the largest relative discrepancy anywhere in the emitted record at 1.3×10^{-11}, on a mean null-site amplitude that accumulates many small terms. That environment ran NumPy 2.0 against the development environment's NumPy 1.26, which is the more useful statement: the agreement is across library generations and not merely across two copies of one installation.

2. Expected output and its uncertainty

The first command writes `found_atoms.csv`, containing for each atom its element, its (x, y, z) in ångström in the reference frame, the per-axis 95% conformal half-widths, the model σ, the species confidence and a flag marking lattice-constrained detections, together with the conformal quantile table. The second writes the Ti–O_6 off-centring with its propagated uncertainty. On the same input the run is deterministic and should reproduce 1834 atoms with precision 0.97, bulk recall 95.6/96.4/95.5% for Pb/Ti/O, species confusion 1.1%, in-plane RMS accuracy 0.032 Å and depth RMS 0.37 Å.

The uncertainty attached to each reported coordinate is the exported conformal half-width, and the statement that validates it is the coverage figure itself: 96% of true errors fall within the nominal 95% interval, per stratum, verified held out on a 50/50 split (Table C2). Typical half-widths are 0.02 Å in-plane and 0.5 Å in depth, rising to 1.5 Å in depth for weak oxygen near the exit surface. Because they are strongly heteroscedastic, the per-atom interval and not a global figure is the quantity to propagate. For the derived polarisation the in-plane off-centring carries a propagated σ of 0.011 Å and the along-beam component 0.237 Å.

3. Tolerances and known failure modes

The single stochastic element is the conformal calibration split, which is seeded; re-seeding changes the per-stratum quantiles by less than 5% and leaves the coverage figures within one percentage point. Coverage is a finite-sample quantity, so strata containing fewer than about twenty atoms, of which the constrained-oxygen entrance band is the example here, show conformal noise of several percentage points and should not be read as failures.

The scripts report their own failure modes rather than absorbing them, and a peer should expect to see both. Approximately 30% of the titanium present do not acquire a complete oxygen cage, and these are excluded from the polarisation map rather than completed from the lattice. A further 5% have a cage of the wrong composition, for which the in-plane error rises to a median of 0.45 Å while the positional interval covers only one of the eight. This second mode is a detection failure rather than a localisation failure, and is visible in the exported cage completeness and species confidence rather than in the coordinate error bars.

Appendix F: The weighted edge and the detection geometry

1. Combining the two oxygen sites

The O K edge of the cell is the multiplicity-weighted sum of one apical and two equatorial oxygen. The two equatorial atoms are related by the four-fold axis and are equivalent for $\mathbf{q} \parallel c$, but not for $\mathbf{q} \perp c$, so the second in-plane orientation was computed explicitly as a third momentum-transfer direction rather than proxied by a similar one. Proxying it by the $\mathbf{q} \parallel c$ spectrum gives 39.1%, and counting the first equatorial atom twice gives 46.4%, against a measured value of 37.1%. Combining sites also requires the core-level shift between them,

[1] https://github.com/tristanmccarthy55/atomfind; `INSTALL.md` is the step-by-step guide and `PEER.md` the protocol, with expected values and tolerances.

which the spectra do not carry, since OptaDOS writes each on a Fermi-referenced axis. Both core-hole calculations share a supercell and hence a ground state, so the difference of their total energies is the final-state shift directly, 0.171 eV, and displacing the equatorial edge by it moves the weighted result from 37.1% to 35.8%. This is well inside the ±1 eV sensitivity sweep that previously stood as the systematic, so the weighted edge carries no free parameter.

2. Convergence and collection averaging

For the O K edge at 300 keV, $\theta_E = 1.09$ mrad and the magic angle is 4.32 mrad $= 3.98\,\theta_E$, reproducing the textbook factor of ≈ 4 and validating the geometry model. Averaging the dipole weights over the convergence and collection cones then gives the numbers behind Sec. III E. At $\beta = 1$ mrad the surviving fraction falls from 0.62 under parallel illumination to 0.29 at a convergence of $\alpha = 2$ mrad and to -0.04 at $\alpha = 5$ mrad, so recovering the full anisotropy would need illumination as parallel as $\theta_E = 1.09$ mrad, which a focused probe cannot provide.

Beyond the magic angle the anisotropy inverts and saturates near a third of its intrinsic value rather than vanishing, and at the $\alpha = 100$ mrad convergence that atomic-resolution ptychography requires it is flat in β, -0.32 to -0.34 from 1 to 100 mrad, the convergence rather than the spectrometer aperture setting the averaging. The 12.1% orientation contrast that leaves needs $\approx 1.2 \times 10^3$ counts per channel for SNR 3, against the 4.5×10^4 implied by a 2% anisotropy, so the hole radius of a hollow detector may be chosen to suit the imaging.

The treatment here is kinematic and dipole-limited, and averages incoherently over incident directions. It carries no thickness dependence, and the channelling check of Sec. III E was run separately; folding both into the aperture average is left as further work.

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